A Novel Token Mixer for Resource Efficient Scientific & Language Foundation Models

A GPU Efficient (Quasi-)Linear Alternative to Quadratic Self-Attention

INVESTMENT THESIS A mathematically grounded trainable token-mixing mechanism is possible that can preserve the familiar transformer block while materially reducing the computation and memory associated with attention. The immediate programme is to validate and scale this architecture in scientific foundation models and medium-scale GPTs.

$\sim 50\%$	$\sim 40\%$	**quasi-linear**
reported reduction in training FLOPs in some tasks	reported reduction in peak memory in some tasks	token-mixing complexity

1 The Opportunity

Foundation models for scientific machine learning increasingly operate over high-resolution spatial grids, while language models must process growing token sequences. Standard self-attention explicitly constructs pairwise token interactions, creating quadratic cost in the number of spatial or language tokens. This limits resolution, context length, batch size and cluster utilisation.

The proposed architecture replaces attention with trainable token-mixing transformations based on multilinear tensor decomposition. Crucially, it preserves the surrounding residual, normalisation and feed-forward structure. This allows controlled, in-place replacement experiments rather than a wholesale redesign of the model and optimisation pipeline.

Self-Attention $\mathcal{O}(n^2)$ token interaction $\Longrightarrow$ **In-Place Replacement** preserve residual, norm and MLP $\Longrightarrow$ **Novel Token Mixing** (quasi-)linear cost

2 Two Mechanisms for the Idea

2.1 A Standard Multilinear Version: Learned Low-Rank Mixing in Every Mode

Each head applies separate learned low-rank transformations along height, width and channel. Every mode matrix is factorised into compression and expansion factors, giving six trainable matrices per head. The layer preserves the input dimension and has linear complexity in that dimension.

This construction can be interpreted as learning the spatial basis rather than fixing it in advance. An FNO is a restricted point of comparison: its spatial factors are fixed to truncated Fourier modes. Learning the factors may increase expressiveness, although it does not inherit the FFT speed or exact translation equivariance supplied by a fixed Fourier basis.

input tensor $H \times W \times C$ $\rightarrow$ height factors compress $\rightarrow$ expand $\rightarrow$ width factors compress $\rightarrow$ expand $\rightarrow$ channel factors compress $\rightarrow$ expand $\rightarrow$ dimension-preserving output

The quantitative reductions reported in this brief are preliminary results and should be treated as evidence to be reproduced and extended in the proposed programme.

2.2 A Spectral Version: Global Mixing with Resolution Compatibility

The spectral candidate uses Fourier bases on periodic domains and DCT-II bases on non-periodic domains. Dense channel mixing is replaced by a sum of independently modulated low-rank heads. The resulting layer has quasi-linear complexity and a global receptive field without constructing a quadratic attention matrix.

A particularly useful property for PDE modeling is resolution compatibility, which the spectral version affords. Relative to an FNO, the construction generalizes channel mixing through multiple independently modulated low-rank components. *But surprisingly this spectral version is also looking most promising in even language tasks!*

3 Evidence to Date

The current evidence comes from matched replacement studies in which every attention layer is replaced while the remaining architecture and optimisation pipeline are held fixed. The reported findings are:

- **Lowering of Training Computation for Standard Tasks:** up to 50% reduction in training FLOPs with almost no deterioration in task performance across standard vision and small data language experiments.
- **Multi-GPU Memory Efficiency for AI-for-Physics:** in solar-physics simulation studies conducted with a NASA group, our mechanism reduced peak memory requirement by 40% while maintaining or improving baseline performance.
- **Strong Performance on Benchmark Language Tasks:** the FLOP requirement and accuracy both improve for standard NLP evaluation of GLUE — particularly so when the optimizer is the modern option of Muon. *This advantage also persist over other attention alternatives like KDA and SSMs*

The present evidence supports a credible efficiency thesis, but the next phase must establish where the standard and spectral variants win, how the gains scale with resolution and model size, and how they compare under a uniform experimental protocol.

4 Why This Matters for GPU-Cluster Utilisation

If the measured FLOP and memory savings persist at larger scale, the same cluster budget can support larger grids, longer contexts, larger batches, more extensive hyperparameter searches or more model replicas. The commercial value is therefore better capability and experimentation per unit of accelerated compute.

Constraint	Attention-Centric Regime	Opportunity from Our Token Mixer
Spatial or Token Scaling	Quadratic pairwise interaction	Linear or quasi-linear token mixing
Peak Memory	Attention activations constrain batch and resolution	Reported 40% reduction in a multi-GPU solar-physics study
Experiment Throughput	Larger runs consume more cluster time	Reported FLOP savings can increase trials per compute budget

5 Next Validation Programme

1. **Structured Design Search.** Do a thorough sweep of head counts, retained modes and low-rank dimensions while measuring task quality, training FLOPs, wall-clock throughput and peak memory — in comparison to basic scientific foundation models.
2. **Resolution and Model Scaling.** Progress to higher-resolution grids and to SOTA scientific foundation-model settings, like of the kind developed by Polymathic AI.
3. **GPT Specific Edits** We believe certain specific "gating" mechanisms are needed in our current map to fully match near-frontier LLMs like GPT2.
4. **Systems Validation.** Profile multi-GPU scaling for communication overhead & kernel efficiency for real scientific use-cases.

6 The Minimal Funding Request

The immediate bottleneck is execution capacity. A rigorous scaling study across scientific foundation models and small GPTs requires *atleast one* dedicated machine-learning engineer or a postdoctoral researcher who can own implementation, benchmarking, distributed training and reproducibility.

Fund One Dedicated Hire
ML engineer or postdoctoral researcher

$\Longrightarrow$

Execute the Scaling Programme
Controlled comparisons (at the compute-optimal/Chinchilla scale) and systems profiling

$\Longrightarrow$

Deliver a Novel SOTA Model
A validated architecture for efficient scientific and language foundation models at $\mathcal{O}(10^2)$M parameter scale.

Role of the Hire

- Implement production-quality standard and spectral token-mixing layers.
- Create matched baselines for attention, FNOs, KDA and state-space alternatives.
- Run systematic searches over heads, modes and ranks.
- Scale training across multiple GPUs and higher-resolution scientific datasets.
- Produce reproducible benchmark artefacts to help wider adoption.

7 Overarching Potential

The programme is not simply an optimisation of a transformer kernel. It tests whether multilinear tensor structure can become a new primitive for global token mixing. Success would create an architecture that combines:

- the familiar optimisation and residual structure of transformers;
- linear or quasi-linear token-mixing complexity;
- global receptive fields in the spectral construction;
- compatibility across admissible spatial resolutions; and
- demonstrable reductions in training computation and memory
  — hence a potential step-change in edge-AI prospects.

THE PROPOSITION We request funding the dedicated technical capacity required to convert promising matched experiments into a decisive scaling study across scientific foundation models and small GPTs. The outcome would be a validated route to an efficient use of GPU clusters via a conceptually distinct alternative to attention.